Lecture 07

Erik An

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1 Sets of Numbers

1.1 Basic Definitions

- **Natural Numbers**: $\mathbb{N} = \{1, 2, 3, \dots\}$

- **Integers**: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- **Rational Numbers**:
$$\mathbb{Q} = \left\{ \frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N} \right\}$$
  - Note: The denominator n cannot be zero ($n \in \mathbb{N}$ ensures this).

1.2 Irreducible Fractions

A fraction $\frac{m}{n}$ is **irreducible** (or in lowest terms) if m and n have no common factor greater than 1.
$$\gcd(m, n) = 1$$

2 Properties of Rational Numbers

2.1 Density of Rational Numbers

Theorem: Between any two rational numbers, there is another rational number.

2.1.1 Proof

Let $a, b \in \mathbb{Q}$ with $a < b$. Consider the midpoint c:
$$c = \frac{a+b}{2}$$
Clearly, $a < c < b$. We must show that $c \in \mathbb{Q}$. Let $a = \frac{m}{n}$ and $b = \frac{p}{q}$ where $m, p \in \mathbb{Z}$ and $n, q \in \mathbb{N}$.
$$c = \frac{1}{2}\left(\frac{m}{n} + \frac{p}{q}\right) = \frac{mq + np}{2nq}$$
Since numerator and denominator are integers (and denominator $\neq 0$), $c \in \mathbb{Q}$.

3 Real and Irrational Numbers

3.1 Existence of Irrational Numbers

Question: Are there other numbers in $\mathbb{R}$ besides $\mathbb{Q}$? **Proposition**: There is a real number x such that $x^2 = 2$. **Definition**: The set of **irrational numbers** is defined as $\mathbb{R} \setminus \mathbb{Q}$.

- Notation: $x = \sqrt{2}$. We previously proved that $\sqrt{2} \notin \mathbb{Q}$.

3.2 Arithmetic of Irrational Numbers

We explore operations between rational and irrational numbers.

1. **Sum of two irrational numbers**: Can be rational.
   - Example: $\sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}$.
2. **Sum of a rational and an irrational number**: Is always irrational.
   - **Claim**: If $a \in \mathbb{Q}$ and $b \notin \mathbb{Q}$, then $a + b \notin \mathbb{Q}$.
   - **Proof (by contradiction)**: Suppose $a + b = x$ where $x \in \mathbb{Q}$. Then $b = x - a$. Since $\mathbb{Q}$ is closed under subtraction, $x - a \in \mathbb{Q}$, which implies $b \in \mathbb{Q}$. This contradicts the assumption that b is irrational.
3. **Product of a non-zero rational and an irrational number**: Is always irrational.
   - **Claim**: If $a \in \mathbb{Q}, a \neq 0$ and $b \notin \mathbb{Q}$, then $a \cdot b \notin \mathbb{Q}$.
   - **Proof**: Similar to the sum proof, if $ab = x \in \mathbb{Q}$, then $b = x/a$. Since rationals are closed under division (for non-zero divisors), b would be rational, a contradiction.

3.3 Density of Irrational Numbers

Theorem: Between any two real numbers, there is an irrational number.

3.3.1 Proof

Let $a, b \in \mathbb{R}$ with $a < b$. We want to find an irrational x such that $a < x < b$. We choose a large enough integer n such that the step size is smaller than the gap $b - a$.

$$n > \frac{\sqrt{2}}{b-a} \implies \frac{\sqrt{2}}{n} < b - a$$

We construct x based on the nature of a:

- **Case 1**: If $a \in \mathbb{Q}$. Consider $x = a + \frac{\sqrt{2}}{n}$. Since a is rational and $\frac{\sqrt{2}}{n}$ is irrational (quotient of irrational and rational), the sum is irrational. Also, $a < a + \frac{\sqrt{2}}{n} < b$ (by our choice of n).
- **Case 2**: If $a \notin \mathbb{Q}$. Consider $x = a + \frac{1}{n}$. Since a is irrational and $\frac{1}{n}$ is rational, the sum is irrational. Also, $a < a + \frac{1}{n} < b$ (assuming n is chosen such that $1/n < b - a$).

Note: The handwritten notes for Case 2 are slightly fragmented but imply adding a rational increment to an irrational base keeps the result irrational.

4 Decimal Representations

4.1 Representation Structure

Any real number x has a decimal representation:

$$x = b_k \dots b_0.a_1 a_2 a_3 \dots$$

$$x = \sum_{i=0}^{k} b_i 10^i + \sum_{i=1}^{\infty} \frac{a_i}{10^i}$$

where digits $a_i \in \{0, 1, \dots, 9\}$.

4.2 Geometric Series and Repeating Decimals

Proposition:

1. Finite Sum: $x + x^2 + \dots + x^n = \frac{x(1-x^n)}{1-x}$ (for $x \neq 1$).
2. Infinite Sum (for $0 < |x| < 1$):

$$S = x(1 + x + x^2 + \dots) \implies S = \frac{x}{1-x}$$

4.2.1 Example: $0.333\dots$

$$0.333\dots = \frac{3}{10} + \frac{3}{100} + \dots = 3\left(\frac{1}{10} + \frac{1}{100} + \dots\right)$$

Using the formula with first term $x = 1/10$:

$$3 \cdot \frac{1/10}{1 - 1/10} = 3 \cdot \frac{1/10}{9/10} = 3 \cdot \frac{1}{9} = \frac{1}{3}$$

4.3 Existence of Decimal Representation

Theorem: Every $x \in (0, 1)$ has a decimal representation.

4.3.1 Construction Logic

1. Divide the interval $[0, 1]$ into 10 sub-intervals of length $1/10$.
2. Find the sub-interval $[\frac{a_1}{10}, \frac{a_1+1}{10}]$ that contains x. This determines the first digit a_1.
$$\frac{a_1}{10} \leq x < \frac{a_1 + 1}{10}$$
3. Subdivide that interval into 10 smaller parts to find a_2, and so on.
$$x = 0.a_1 a_2 \ldots$$

4.4 Positional Notation Examples

- $12 = 1 \cdot 10^1 + 2 \cdot 10^0$
- $12.5 = 1 \cdot 10^1 + 2 \cdot 10^0 + 5 \cdot 10^{-1}$
- Notation varies: 12.5 (dot) vs $12,5$ (comma).
- Approximation: $\sqrt{2} \approx 1.4142\ldots$